

FAN COILS ACCESSORY ELECTRIC HEATERS

WIRING DIAGRAMS

FIG.	FIELD- INSTALLED HEATER MODEL	FB4C	FE4A/FE5A	FH4C	FV4C	FX4D	FY5B	FZ4A	PF4MA	PF4MB	LABEL
1	KFCEH0401N03A	18,24	x	001	x	19,25	18,24	24	18,19,24,25	x	340816-101
1	KFCEH0501N05A	18-60	002-006	001-002	002-006	19-61	18-60	24-61	18-61	19-61	340816-101
2	KFCEH0801N08A	18-60	002-006	001-003	002-006	19-61	18-60	24-61	18-61	19-61	340817-101
2	KFCEH0901N10A	18-60	002-006	001-004	002-006	19-61	18-60	24-61	18-61	19-61	340817-101
6	KFCEH1601315A	42-60	002-006	001-004	002-006	43-61	18-60	48-61	18-61	19-61	340819-101
7	KFCEH2001318A	42-60	003-006	001-004	002-006	43-61	42-60	48-61	42-61	37-61	340818-101
1	KFCEH2401C05A	18-60	002-006	001-002	002-006	19-61	18-60	24-61	18-61	19-61	340816-101
2	KFCEH2501C08A	18-60	002-006	001-003	002-006	19-61	18-60	24-61	18-61	19-61	340817-101
2	KFCEH2601C10A	18-60	002-006	001-004	002-006	19-61	18-60	24-61	18-61	19-61	340817-101
3	KFCEH2901N09A	36-60	002-006	003-004	002-006	37-61	36-60	36-61	36-60	31-61	340813-101
4	KFCEH3001F15A	24-60	002-006	001-004	002-006	25-61	24-60	24-61	24-61	19-61	340814-101
4	KFCEH3101C15A	24-60	002-006	001-004	002-006	25-61	24-60	24-61	24-61	19-61	340814-101
5	KFCEH3201F20A	30-60	002-006	002-004	002-006	31-61	30-60	36-61	30-61	19-61	340815-101
5	KFCEH3301C20A	30-60	002-006	002-004	002-006	31-61	30-60	36-61	30-61	19-61	340815-101
8,9	KFCEH3401F24A	48,60	004-006	003-004	005-006	49-61	48-60	48-61	48-61	49-61	340821-101, 340820-101
8,9	KFCEH3501F30A	48,60	004-006	003-004	005-006	49-61	48-60	48-61	48-61	49-61	340821-101 340820-101

FAN COIL WITH RBC X-13 MOTOR OR BROAD OCEAN DIGI MOTOR

FIG.	FACTORY-INSTALLED HEATER MODEL	FB4C	FX4D	FZ4A	LABEL
11	MKFCEH0501N05A	18,24	19,25	--	341018-101
10	MKFCEH0801N08A	18,30	31	--	341017-101
10	MKFCEH0901N10A	24-48	37-61	--	
12	MKFCEH3001F15A	36-60	61	--	341019-101
29	MKFCEH0801N08A	--	--	36,48	342416-101
	MKFCEH0901N10A	--	--		
30	MKFCEH3001F15A	--	--	60,61	342418-101
31	MKFCEH0501N05A	--	--	24	342417-101

FAN COIL WITH COOLING ONLY CONTROL

FIG.	MODEL	SIZE	LABEL
13	FV4C	002-006	326014-101
14	FE4A/FE5A	002-006	333107-101
15	FY5B/PF4MNA	18-60	328964-101
15	FH4C	001-004	328964-101
16	FB4C/FX4D (RBC)	18-61	336228-101
16	PF4MNA/B	19,25,31,37,43,49,61	336228-101
17	FB4C/FX4D (BOM)	18-61	337519-101
32	FZ4A	24-61	342415-101

ELECTRIC HEATERS

FIG.	HEATER MODEL	FF1E	LABEL
18	KFDEH0801D05A	18,24,30,36	341080-101
19	KFDEH0901D75A	18,24,30,36	341081-101
19	KFDEH1001D11A	18,24,30,36	341081-101
20	KFEEH0101D05A	19,25,31,37	341082-101
21	KFEEH0201D75A	19,25,31,37	341083-101
21	KFEEH0301D11A	19,25,31,37	341083-101

FIG.	HEATER MODEL	CONTROL TYPE	FFMA	LABEL
22	EHK2-05B	Sequencer - HS	18,24,30,36 Prior to serial number date code 1715V.	202070290385
22	EHK2-08B	Sequencer - HS	18,24,30,36 Prior to serial number date code 1715V.	202070290385
22	EHK2-10B	Sequencer - HS	18,24,30,36 Prior to serial number date code 1715V.	202070290385
23			18-37 Serial number date code 1715V and later.	2020702A717
24	EHK2-05B	Relay - HR	18-37	06-7094-02
24	EHK2-08B	Relay - HR	18-37	06-7094-02
24	EHK2-10B	Relay - HR	18-37	06-7094-02

FIG.	FFMA	LABEL
25	19,31	--
26	25,37	--

FIG.	FPM(A,B)N(C,U)	LABEL
27	ALL	202070290388 Valid for models FPM(A,B)N(C,U)0**000AAAA
27	FPMA with Time Delay Board	2020702A2016 Valid for models FPM(A,B)N(C,U)0**000ABAA

FIG.	HEATER MODEL	FPMAN(C,U)	FPMBN(C,U)	LABEL
28	EHK3-05B	18-36	18-30	06-7094-03
28	EHK3-08B	18-36	18-30	06-7094-03
28	EHK3-10B	18-36	18-30	06-7094-03

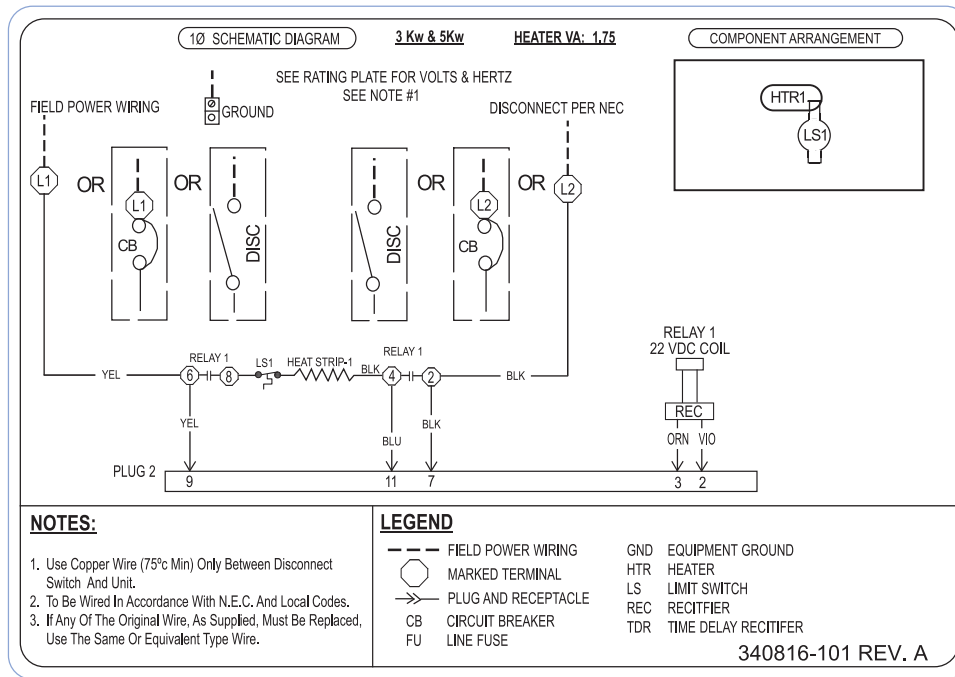


Fig. 1 - 340816-101

A150105

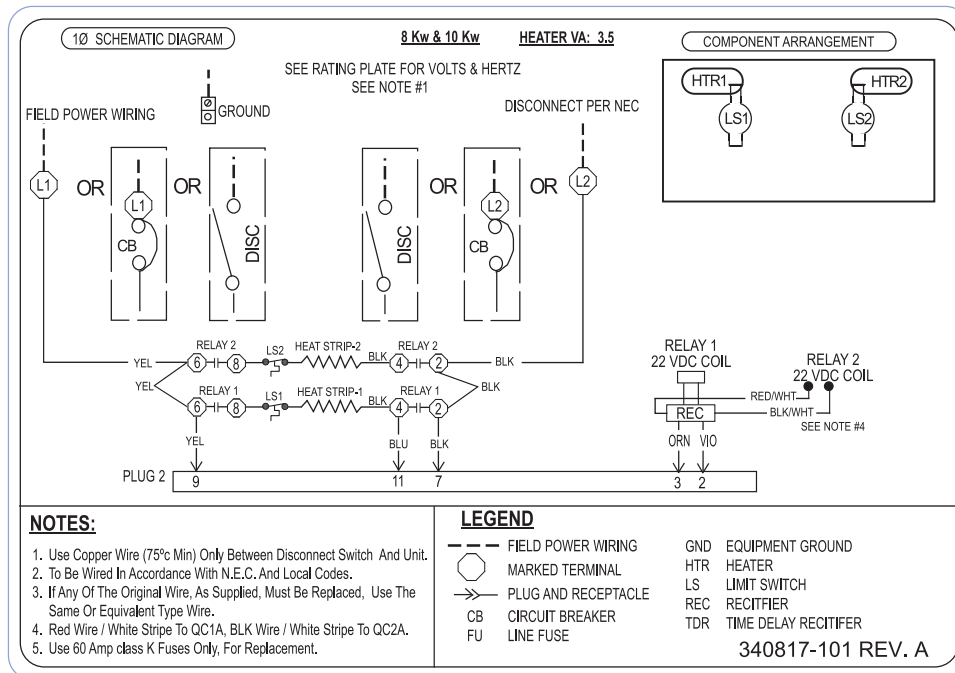


Fig. 2 - 340817-101

A150106

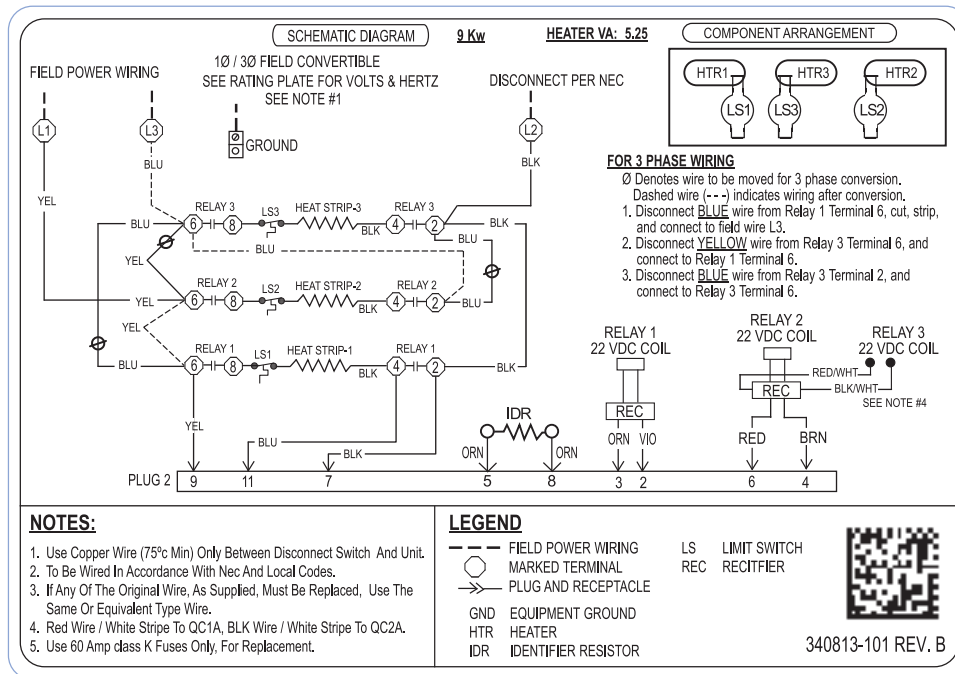


Fig. 3 - 340813-101

A150102

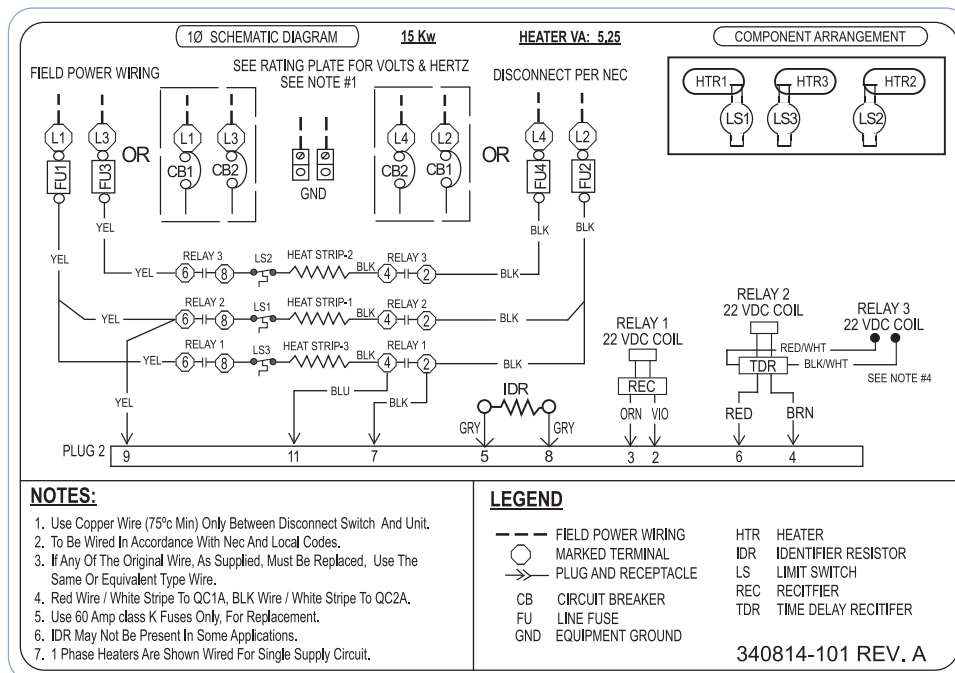


Fig. 4 - 340814-101

A150103

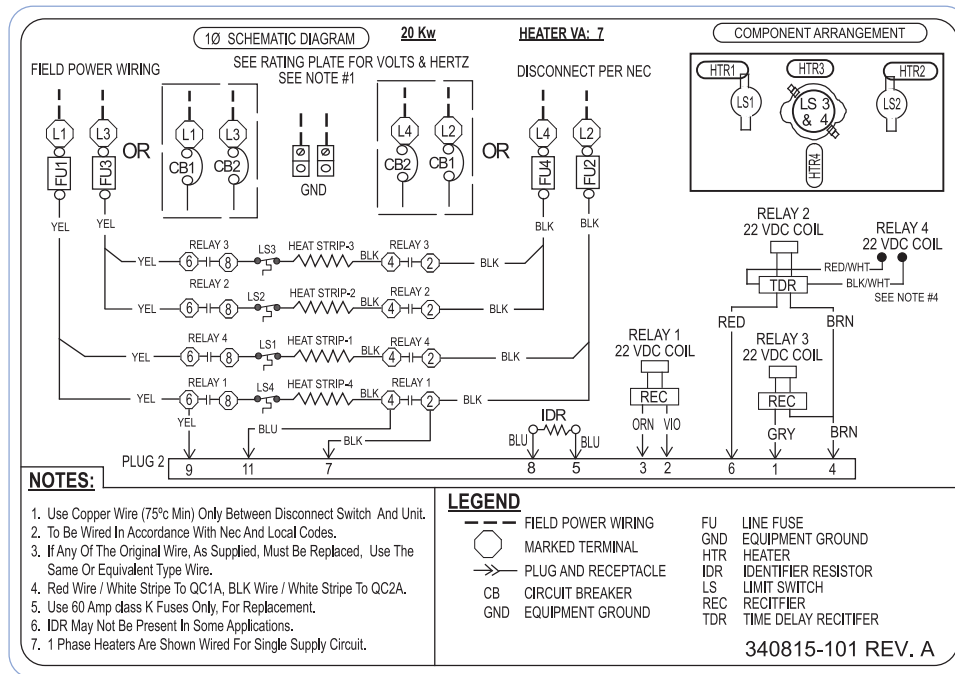


Fig. 5 - 340815-101

A150104

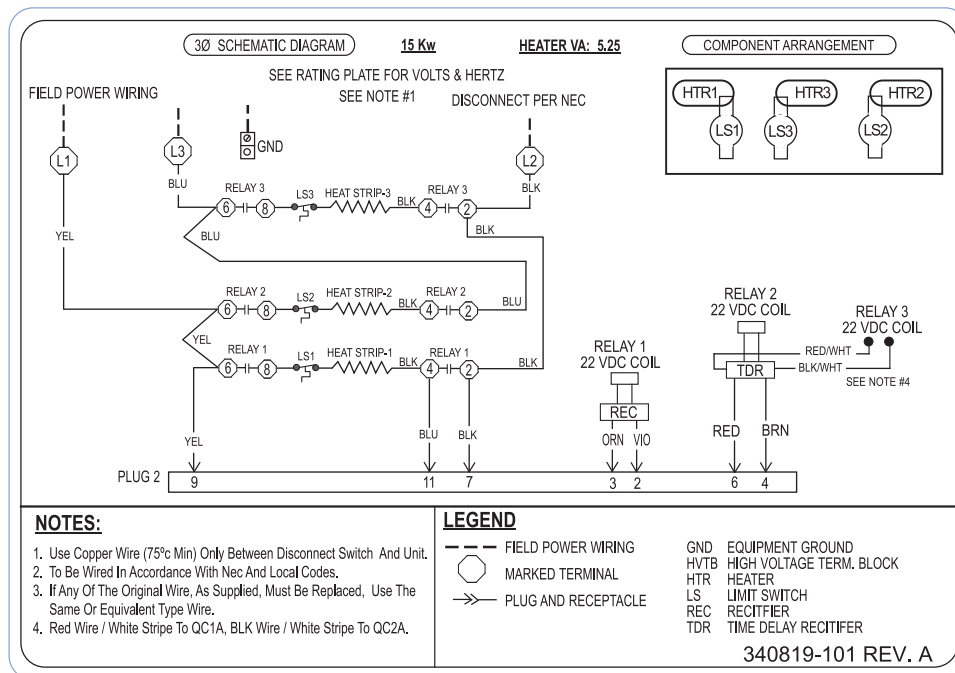


Fig. 6 - 340819-101

A150108

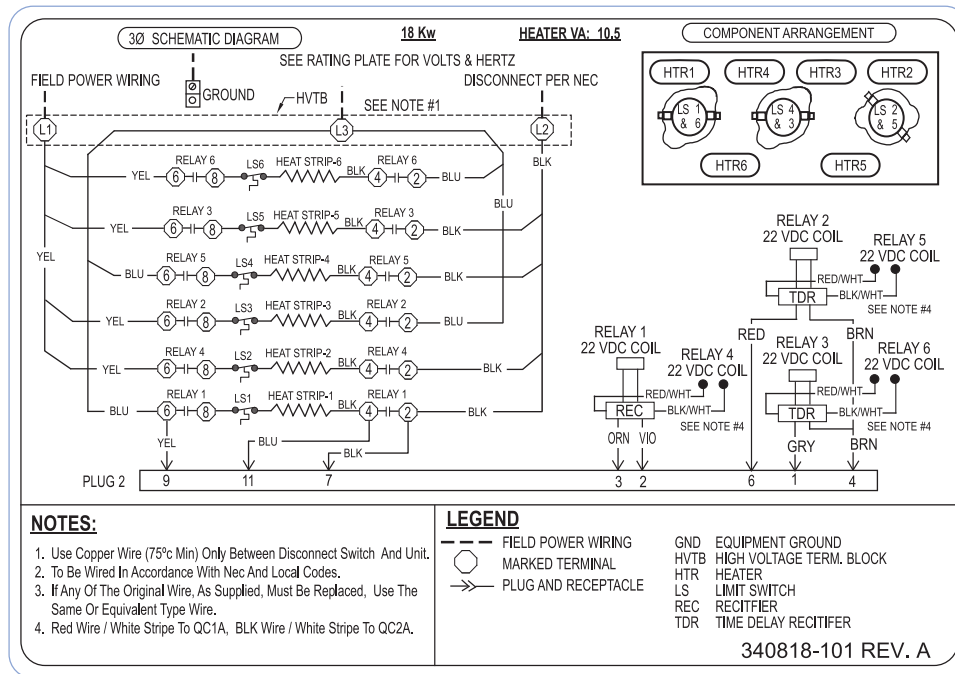


Fig. 7 - 340818-101

A150107

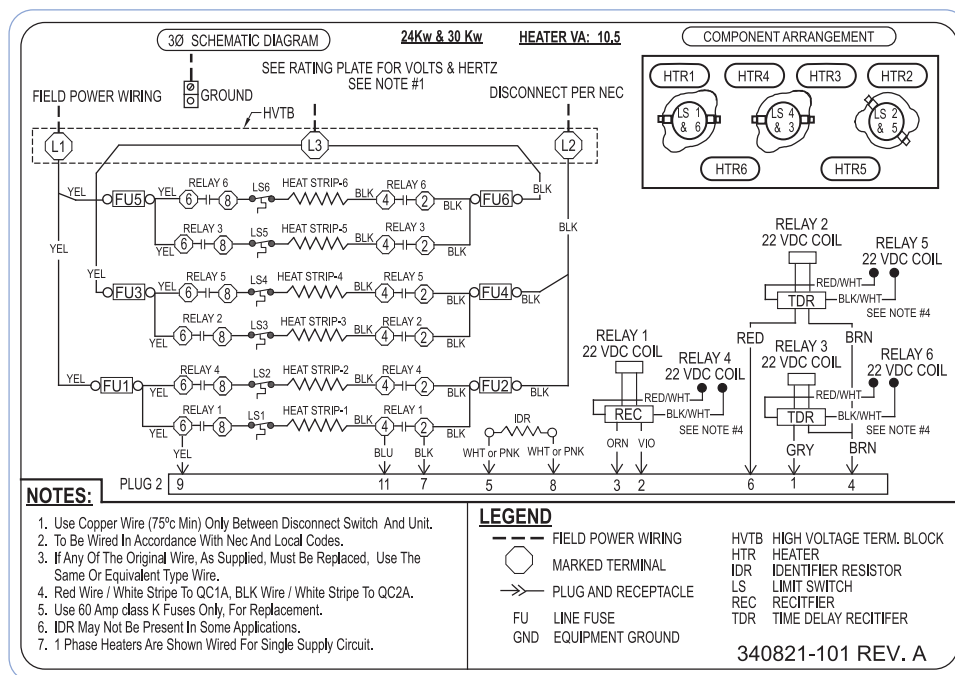


Fig. 8 - 340821-101

A150109

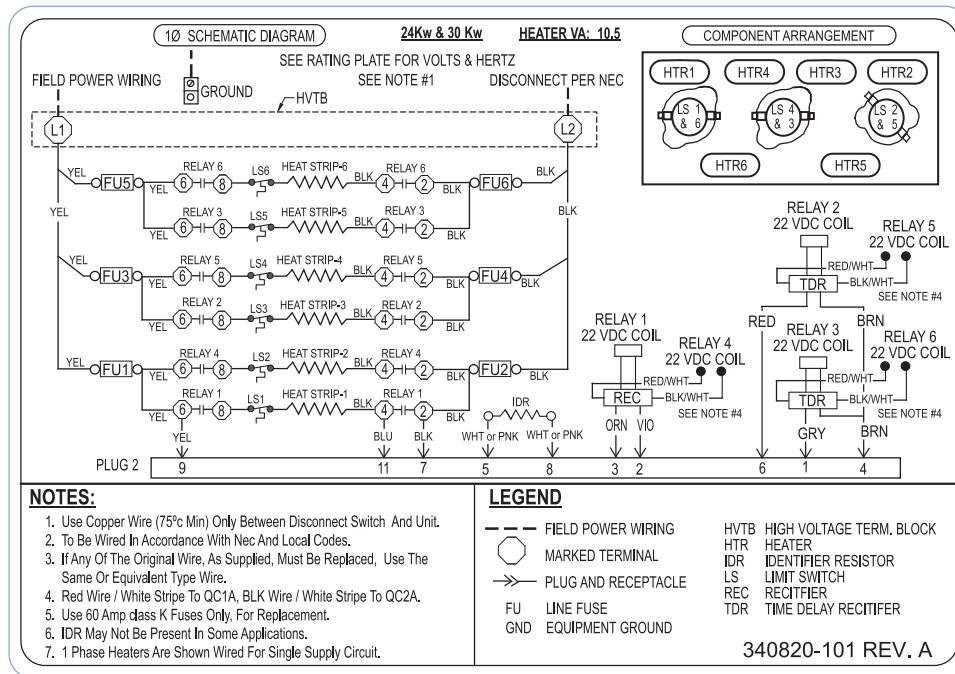


Fig. 9 - 340820-101

A150110

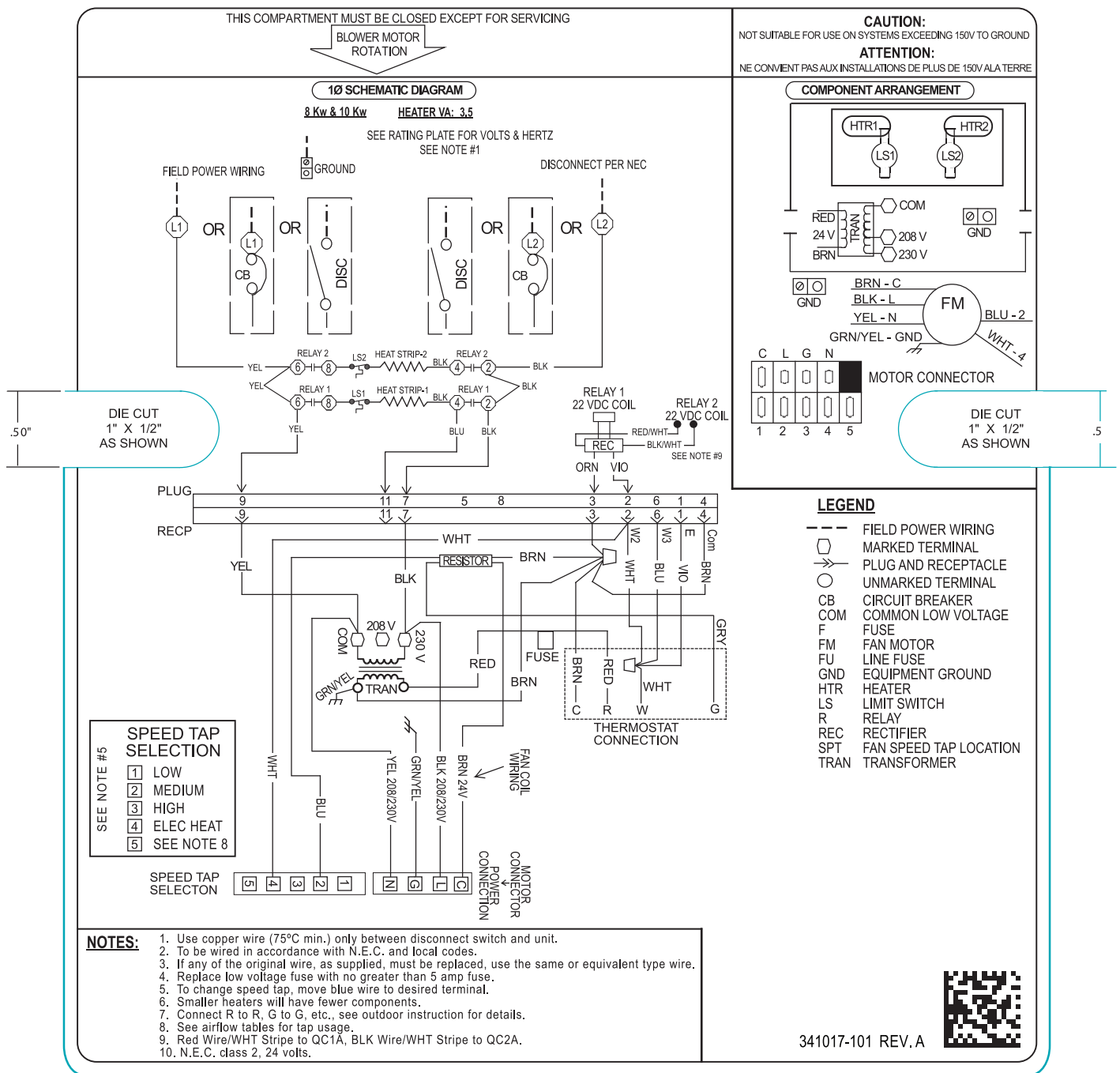


Fig. 10 - 341017-101

A150111

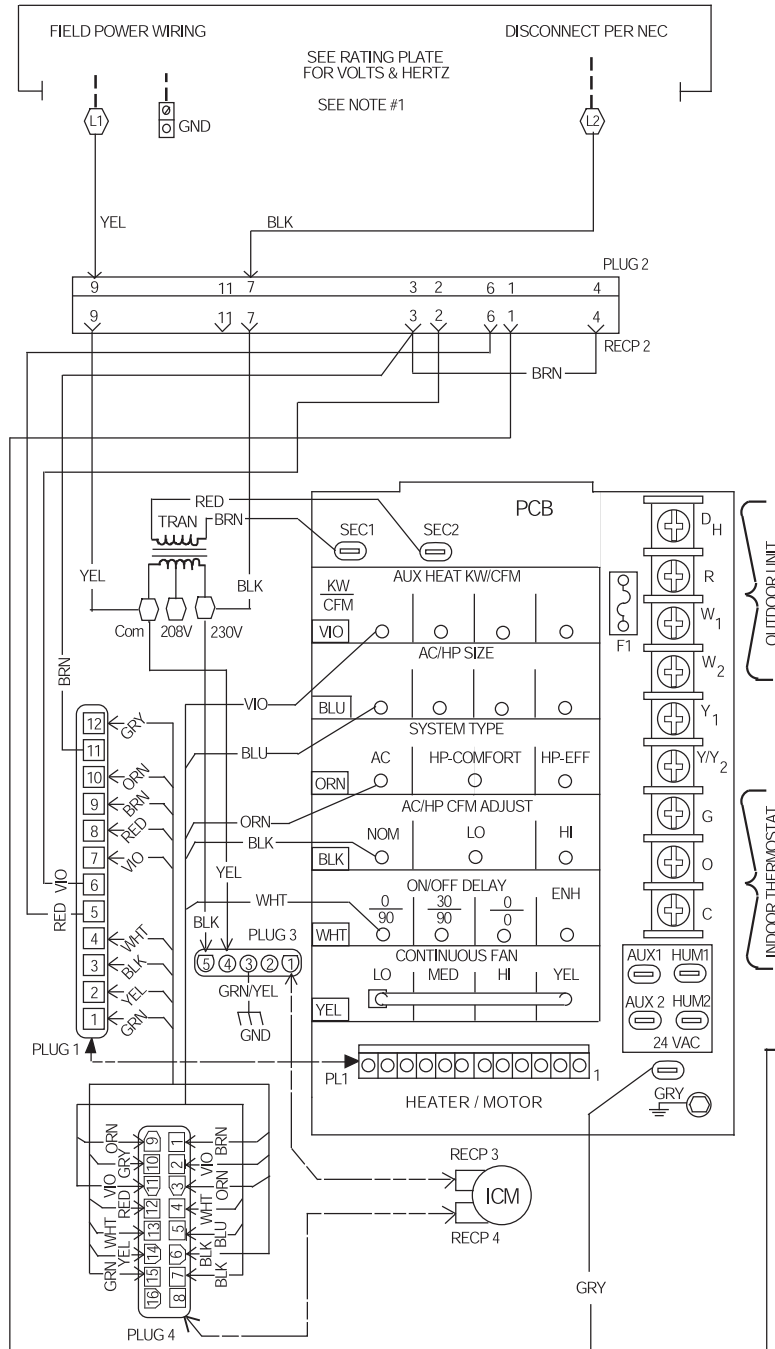


A150112

**THIS COMPARTMENT MUST BE
CLOSED EXCEPT FOR SERVICING**

**BLOWER MOTOR
ROTATION**

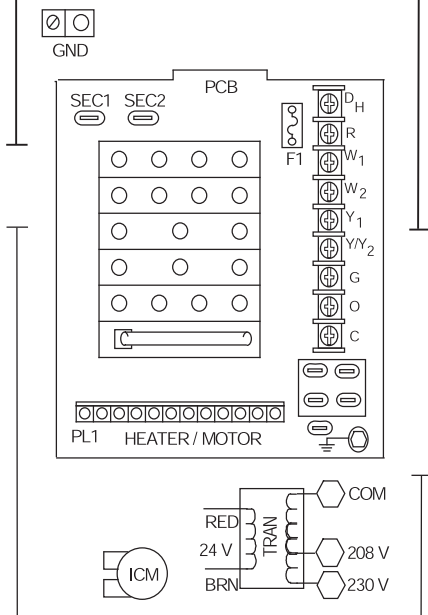
COOLING ONLY SCHEMATIC DIAGRAM



LEGEND

---	FIELD POWER WIRING
⬢	MARKED TERMINAL
⌋⌋	PLUG AND RECEPTACLE
COM	COMMON
F1	LOW VOLTAGE FUSE
GND	EQUIPMENT GROUND
ICM	FAN MOTOR
PCB	PRINTED CIRCUIT BOARD
RECP	RECEPTACLE
TRAN	TRANSFORMER

COMPONENT ARRANGEMENT



NOTES:

1. USE COPPER WIRE (75°C MIN) ONLY BETWEEN DISCONNECT SWITCH AND UNIT.
2. TO BE WIRED IN ACCORDANCE WITH N.E.C. AND LOCAL CODES.
3. TRANSFORMER PRIMARY LEADS, BLUE 208V, RED 230V.
4. IF ANY OF THE ORIGINAL WIRE, AS SUPPLIED, MUST BE REPLACED, USE THE SAME OR EQUIVALENT TYPE WIRE.
5. REPLACE LOW VOLTAGE FUSE WITH NO GREATER THAN 5 AMP FUSE.
7. USE 60 AMP CLASS K FUSES ONLY, FOR REPLACEMENT.
8. CONNECT R TO R, G TO G, ETC., SEE OUTDOOR INSTRUCTION FOR DETAILS.

326014-101 REV. D

Fig. 13 - 326014-101

A07029

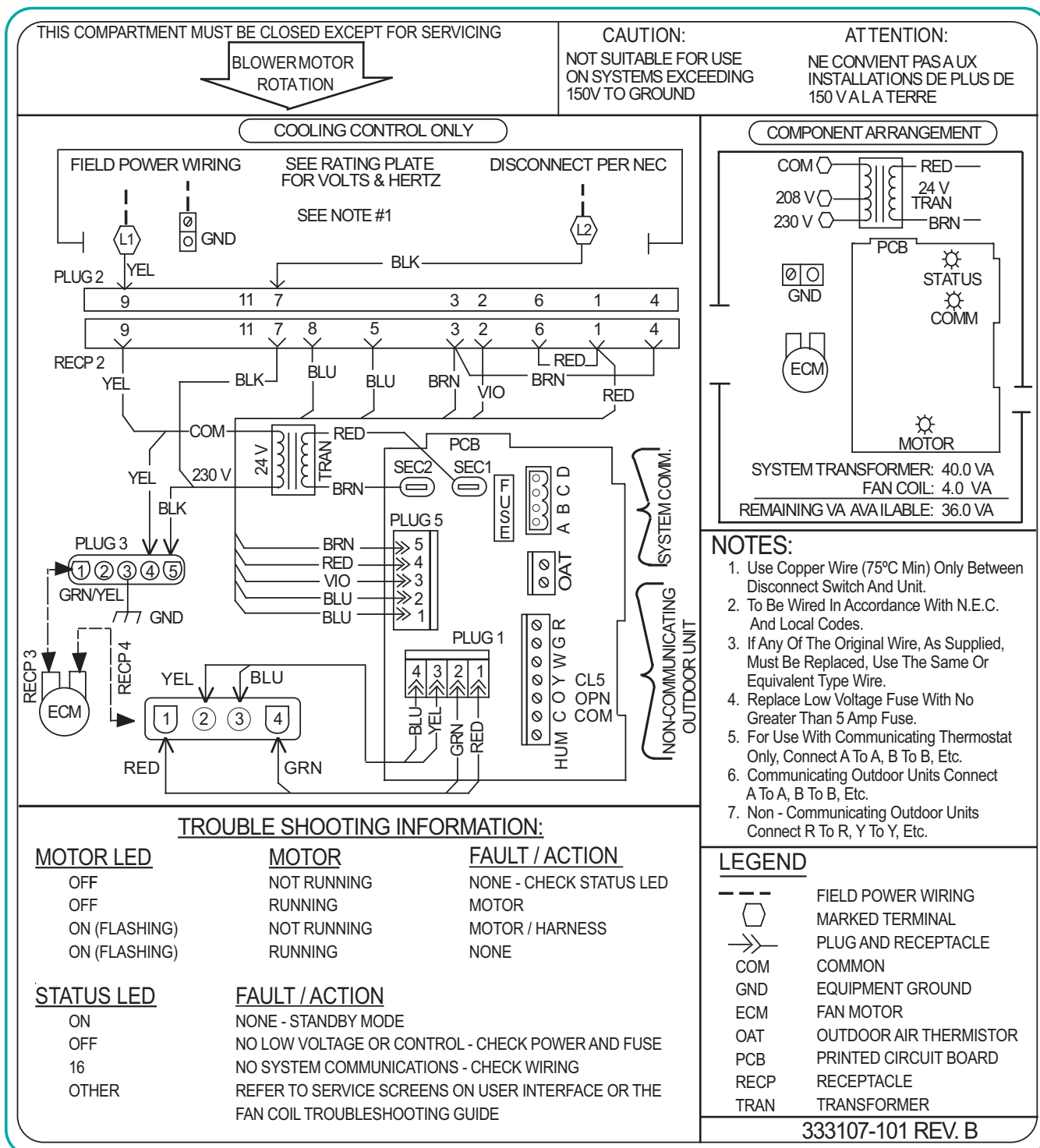
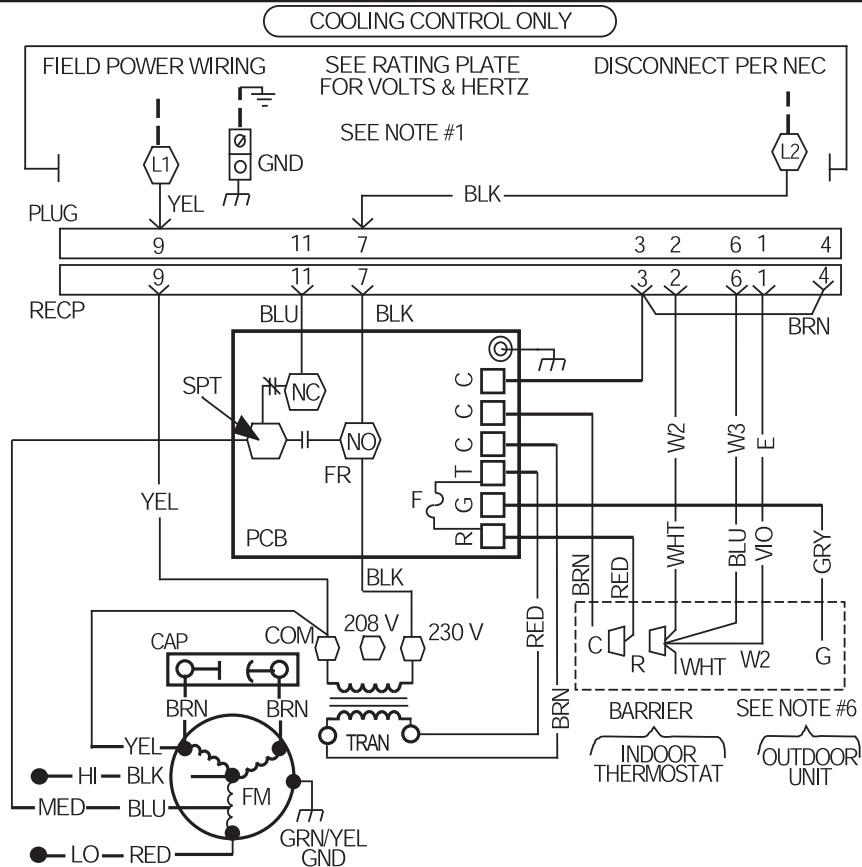


Fig. 14 - 333107-101

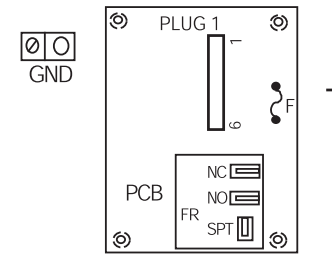
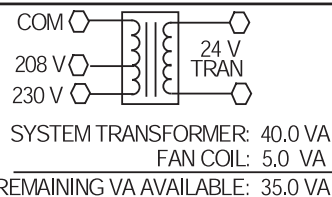
A12363

THIS COMPARTMENT MUST BE CLOSED EXCEPT FOR SERVICING

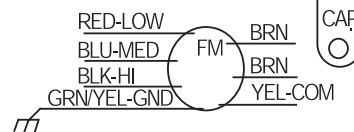
BLOWER MOTOR
ROTATION



COMPONENT ARRANGEMENT



SEE NOTE #5



FAN MOTOR THERMALLY PROTECTED

LEGEND

CAP	CAPACITOR
COM	COMMON
F	LOW VOLTAGE FUSE
FR	PCB FAN RELAY
FM	FAN MOTOR
GND	EQUIPMENT GROUND
PCB	PRINTED CIRCUIT BOARD
RECP	RECEPTACLE
SPT	FAN SPEED TAP LOCATION
TRAN	TRANSFORMER

○	UNMARKED TERMINAL
---	FIELD POWER WIRING
◻	MARKED TERMINAL
→→	PLUG AND RECEPTACLE

328964-101 REV. A

Minimum Motor Speed Tap Selection For Electric Heater

MODEL SIZE	HEATER SIZE KW						
	3,5,8	9	10	15	18	20	24, 30
18	MED *	----	HI	----	----	----	----
24	MED ‡	----	MED ‡	MED ‡	----	---	----
30	----	----	LO	LO	----	MED *	----
36, 42, 48, 60	----	LO	LO	LO	LO	LO	LO
70	----	MED	MED	MED	MED	MED	MED

* - MED speed on 3 speed motors
and HI speed on 2 speed motors.
‡ - MED speed on 3 speed motors
and LO speed on 2 speed motors.

NOTES

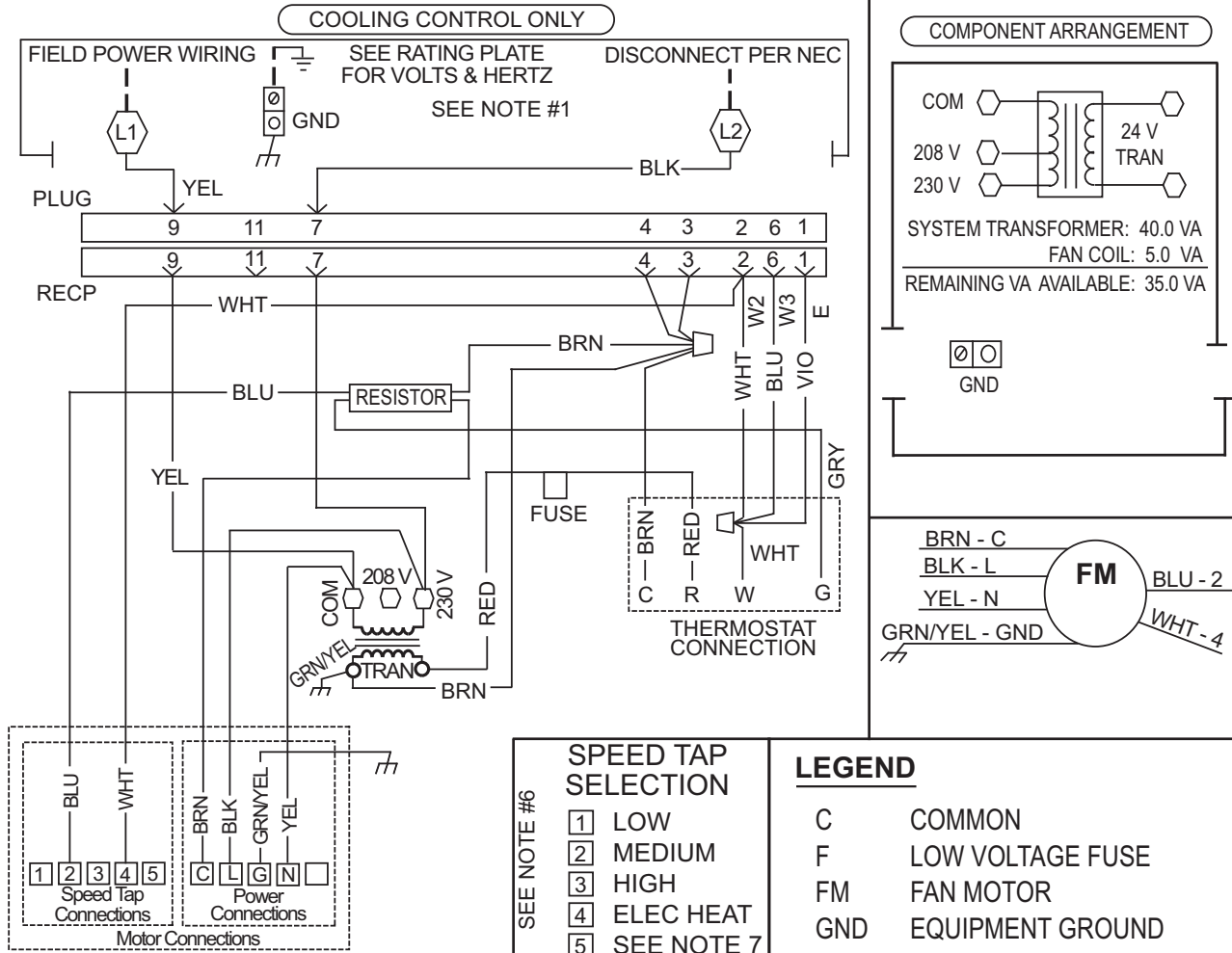
1. Use Copper Wire (75°C Min) Only Between Disconnect Switch And Unit.
2. To Be Wired In Accordance With NEC And Local Codes.
3. If Any Of The Original Wire, As Supplied, Must Be Replaced, Use The Same Or Equivalent Type Wire.
4. Replace Low Voltage Fuse With No Greater Than 5 Amp Fuse.
5. (3) Speed Motor Shown Optional (2) Speed Motor Uses HI (BLK) And LOW (BLUE or RED).
6. Connect R To R, G To G, Etc. See Outdoor Instruction For Details.

Fig. 15 - 328964-101

A07027

THIS COMPARTMENT MUST BE CLOSED EXCEPT FOR SERVICING

BLOWER MOTOR
ROTATION



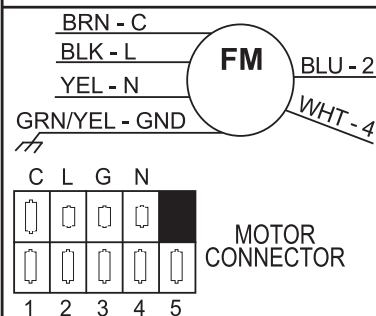
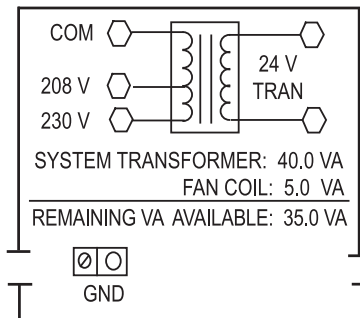
NOTES

1. Use Copper Wire (75°C Min) Only Between Disconnect Switch And Unit.
2. To Be Wired In Accordance With NEC And Local Codes.
3. If Any Of The Original Wire, As Supplied, Must Be Replaced. Use The Same Or Equivalent Type Wire.
4. Replace Low Voltage Fuse With No Greater Than 5 Amp Fuse.
5. Connect R To R, G To G, Etc. See Outdoor Instruction For Details.
6. To change speed tap, move blue wire to desired terminal.
7. See airflow tables for tap usage.

Fig. 16 - 336228-101

A12364

BLOWER MOTOR ROTATION



C	COMMON
F	LOW VOLTAGE FUSE
FM	FAN MOTOR
GND	EQUIPMENT GROUND
RECP	RECEPTACLE
SPT	FAN SPEED TAP LOCATION
TRAN	TRANSFORMER
○	UNMARKED TERMINAL
---	FIELD POWER WIRING
⬡	MARKED TERMINAL
⌋⌋	PLUG AND RECEPTACLE

1. Use Copper Wire (75°C Min) Only Between Disconnect Switch And Unit.
2. To Be Wired In Accordance With NEC And Local Codes.
3. If Any Of The Original Wire, As Supplied, Must Be Replaced. Use The Same Or Equivalent Type Wire.
4. Replace Low Voltage Fuse With No Greater Than 5 Amp Fuse.
5. Connect R To R, G To G, Etc. See Outdoor Instruction For Details.
6. To Change Speed Tap, Move Blue Wire To Desired Terminal.
7. See Airflow Tables For Tap Usage.
8. Factory wires may be present. DO NOT USE

337519-101 REV. B



A12365

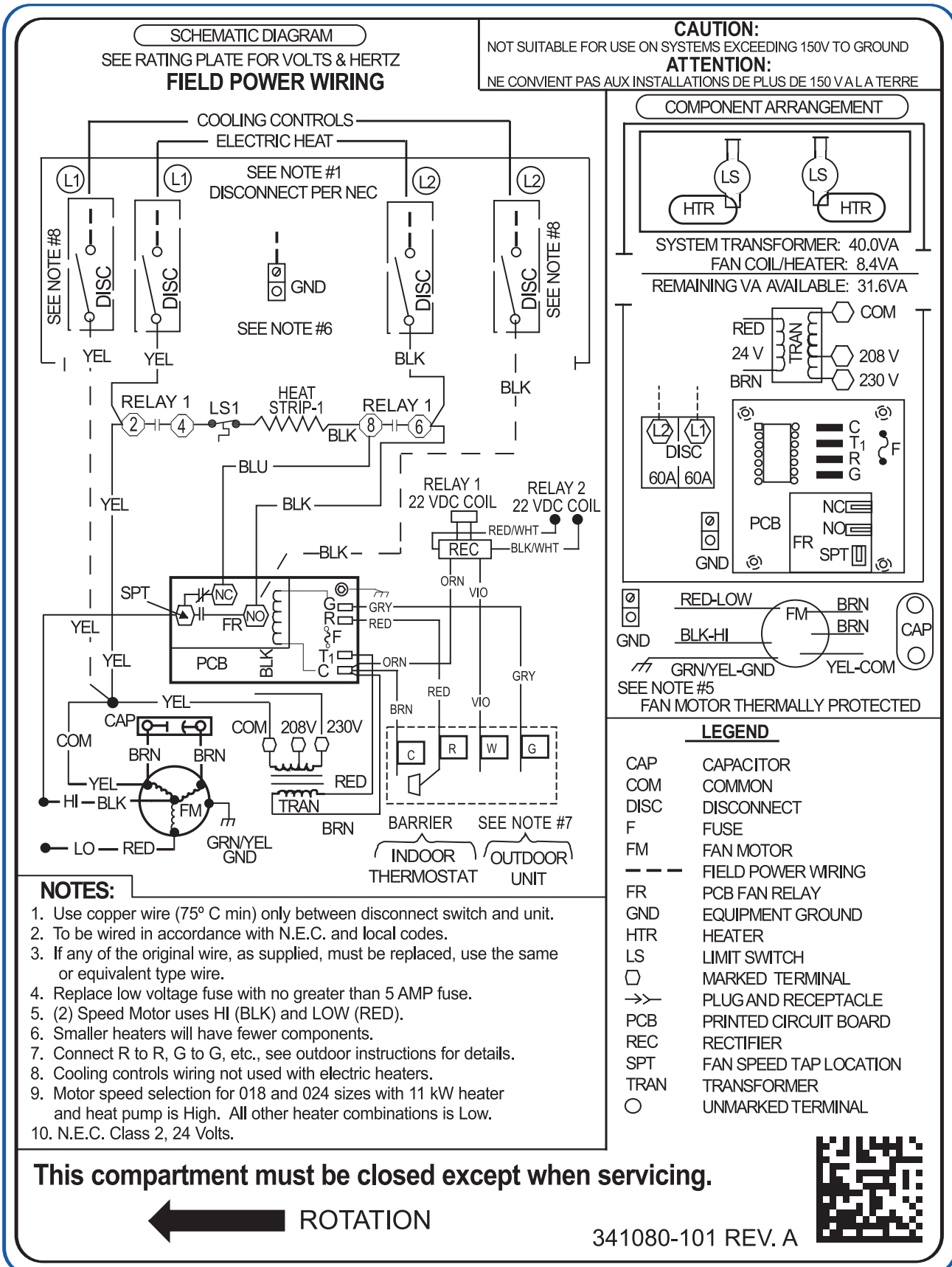


Fig. 18 - 341080-101

A150114

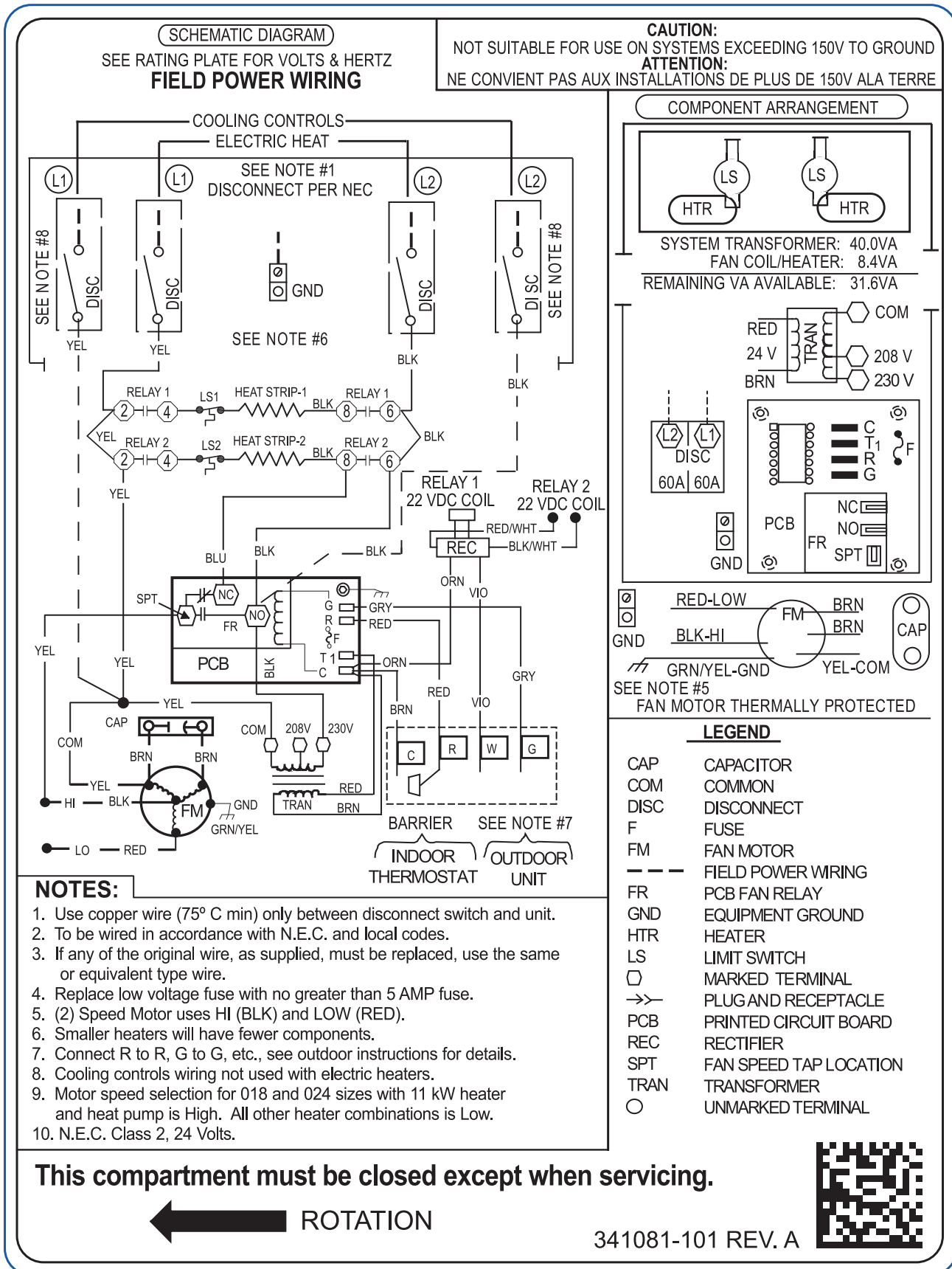


Fig. 19 - 341081-101

A150115

SEE RATING PLATE FOR VOLTS & HERTZ
FIELD POWER WIRING



1. Use copper wire (75° C min) only between disconnect switch and unit.
2. To be wired in accordance with N.E.C. and local codes.
3. If any of the original wire, as supplied, must be replaced, use the same or equivalent type wire.
4. Replace low voltage fuse with no greater than 3 AMP fuse.
5. Smaller heaters will have fewer components.
6. Connect R to R, G to G, etc., see outdoor instructions for details.
7. Cooling speed selection can be tap 1, 2, 3, or 5.
8. Heating speed selection must be tap 4 only.
9. N.E.C. class 2, 24 volts.

	FIELD POWER WIRING
	MARKED TERMINAL
	PLUG AND RECEPTACLE
	UNMARKED TERMINAL
	WIRE NUT
COM	COMMON
DISC	DISCONNECT
F	FUSE
FM	FAN MOTOR
GND	EQUIPMENT GROUND
HIR	HEATER
LS	LIMIT SWITCH
R	RESISTOR
REC	RECTIFIER
SPT	FAN SPEED TAP LOCATION
TB	TERMINAL BLOCK
TRAN	TRANSFORMER

This compartment must be closed except when servicing.



ROTATION

341082-101 REV. A



Fig. 20 - 341082-101

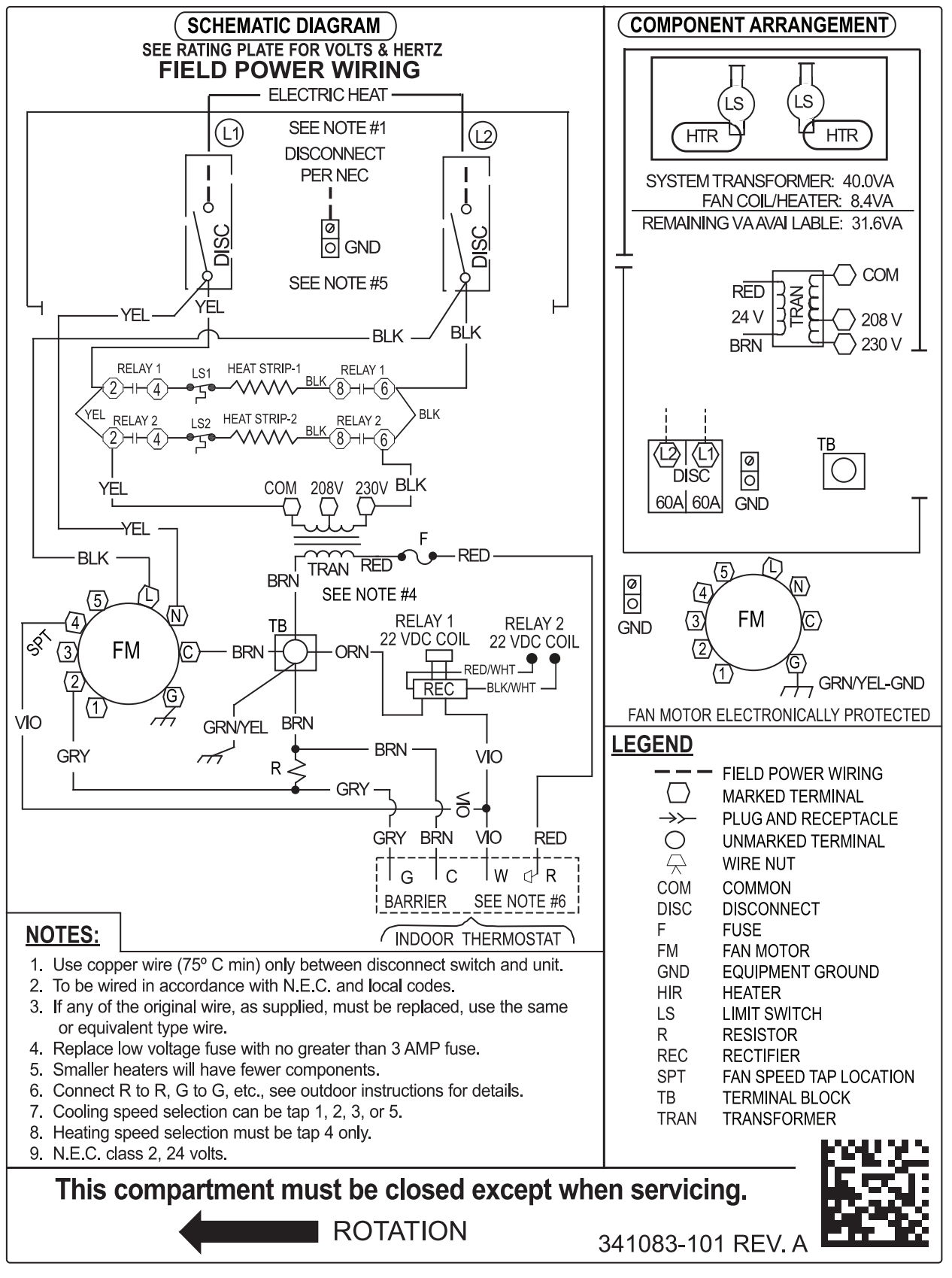
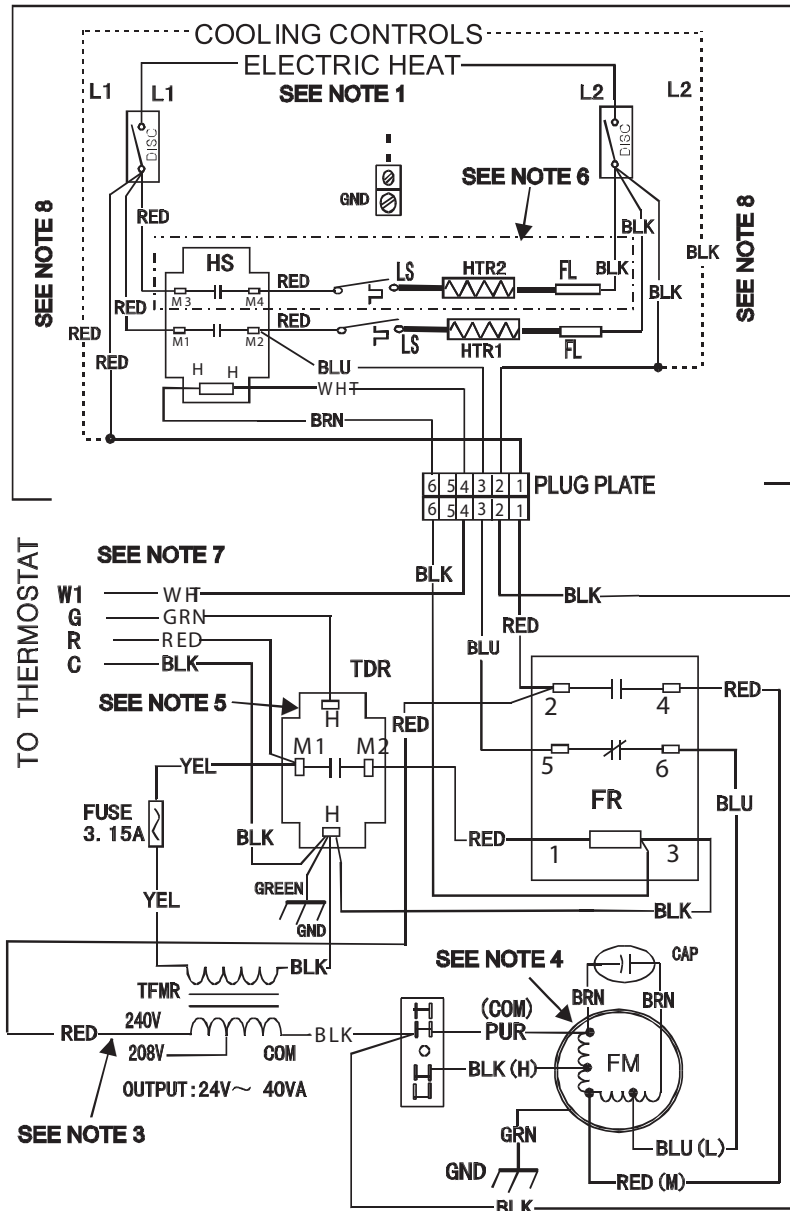


Fig. 21 - 341083-101

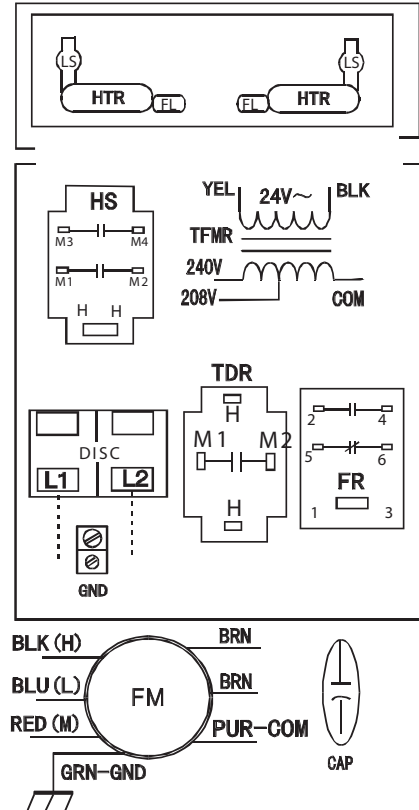
A150117

SCHEMATIC DIAGRAM
SEE RATING PLATE FOR VOLTS&HERTZ
FIELD POWER WIRING

CAUTION:
NOT SUITABLE FOR USE ON SYSTEMS EXCEEDING 150V TO GROUND
ATTENTION:
NE CONVIENT PAS AUX INSTALLATIONS DE PLUS DE 150V A LA TERRE



COMPONENT ARRANGEMENT



DISC DISCONNECT
FR FAN RELAY
HS HEAT SEQUENCER
LS LIMIT SWITCH
TDR TIME DELAY RELAY
TFMR TRANSFORMER
FL FUSE LINK
FM FAN MOTOR
CAP FAN CAPACITOR
GND GROUND
... FIELD POWER WIRING

NOTES:

- 1: Use copper wire(75°C min) only between disconnect switch and unit, To be wired in accordance with N.E.C. and local codes.
- 2: If any of the original wire as supplied must be replaced, use the same or equivalent type wire.
- 3: Remove the red lead from "240V" terminal and then connect the red lead to "208V" terminal on the transformer for 208 volts.
- 4: Factory default fan speed is medium; FM red wire connected to FR #4. For HI speed connect FM black wire to FR #4. For LOW speed connect FM blue wire to FR #4 and FM red wire to FR #6. Always connect the unused FM wire to the dummy terminal block.
- 5: TDR has a 1-30s on delay when "G" is energized and a 45-75s off delay when "G" is de-energized.
- 6: The 5kW heater kit has HTR1 only. Fan coils equipped with electric heat connect power supply to circuit breaker.
- 7: Connect R to R, G to G, etc. See outdoor or indoor instructions for details.
- 8: Cooling controls wiring not used with electric heaters.

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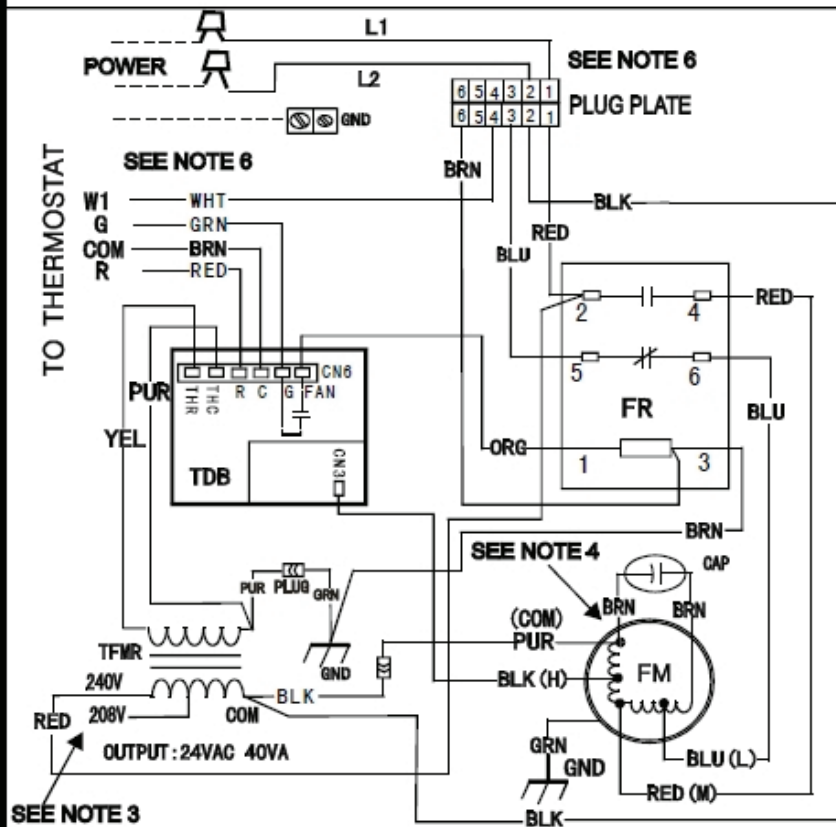
Fig. 22 - FFMANP(018,024,030,036 and EHK2 Electric Heaters with Sequencers
NOTE: Representative for FFMANP(018,024,030,036) prior to serial number date code 1715V

A13131

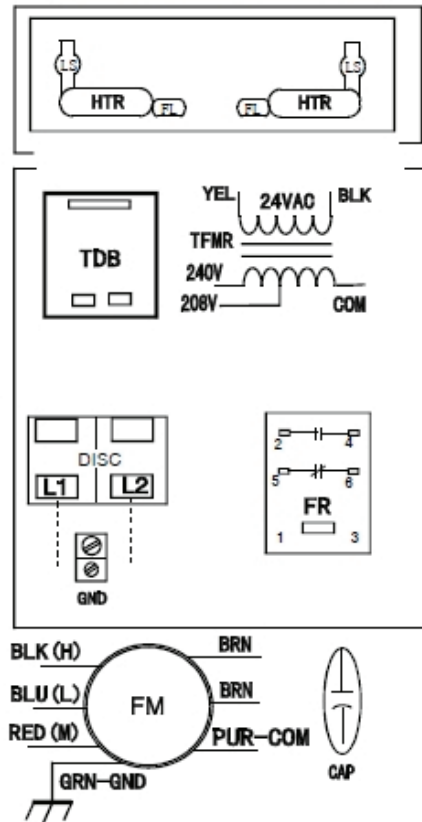
SCHEMATIC DIAGRAM
SEE RATING PLATE FOR VOLTS&HERTZ

CAUTION:
NOT SUITABLE FOR USE ON SYSTEMS EXCEEDING 150V TO GROUND
ATTENTION:
NE CONVIENT PAS AUX INSTALLATIONS DE PLUS DE 150V A LA TERRE

ELECTRIC HEAT WIRING CONNECTION (WHEN APPLIED)



COMPONENT ARRANGEMENT



FR FAN RELAY
TDB TIME DELAY BOARD
TFMR TRANSFORMER
FM FAN MOTOR
CAP FAN CAPACITOR
GND GROUND
- - - FIELD POWER WIRING

NOTES:

- 1: Use copper wire (75°C min), to be wired in accordance with N.E.C. and local codes.
- 2: If any of the original wire as supplied must be replaced, use the same or equivalent type wire.
- 3: Remove the red lead from "240V" terminal and then connect the red lead to "208V" terminal on the transformer for 208 volts.
- 4: Factory default fan speed is medium; FM red wire connected to FR #4. For HI speed connect FM black wire to FR #4. For LOW speed connect FM blue wire to FR #4 and FM red wire to FR #6. Always connect the unused FM wire to the dummy terminal CN3.
- 5: TDB has a 90-100s off delay when "G" is de-energized.
- 6: Connect R to R, G to G, etc. See outdoor or indoor instructions for details.
- 7: Cooling controls wiring not used with electric heaters, connect the plug to electric heaters kit when applied.
- 8: N.E.C., class 2, 24 volts.

2020702A1717

Fig. 23 - FFMANP(018,024,030,036 with PCB Time Delay

NOTE: Representative for FFMANP(018,024,030,036) serial number date code 1715V and later.

A150153

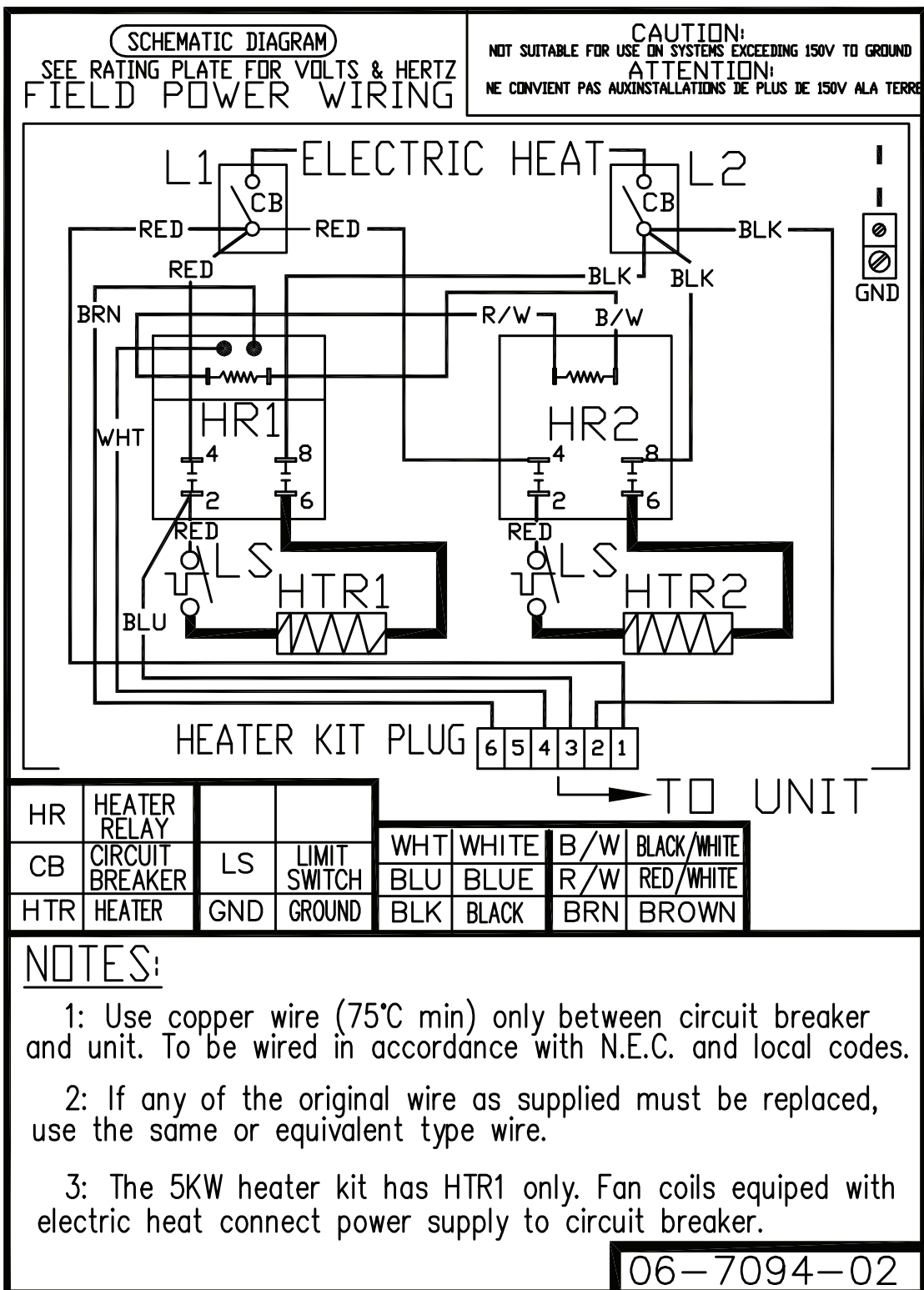


Fig. 24 - EHK2 With Heater Relays

A150100

SCHEMATIC DIAGRAM

SEE RATING PLATE FOR VOLTS&HERTZ

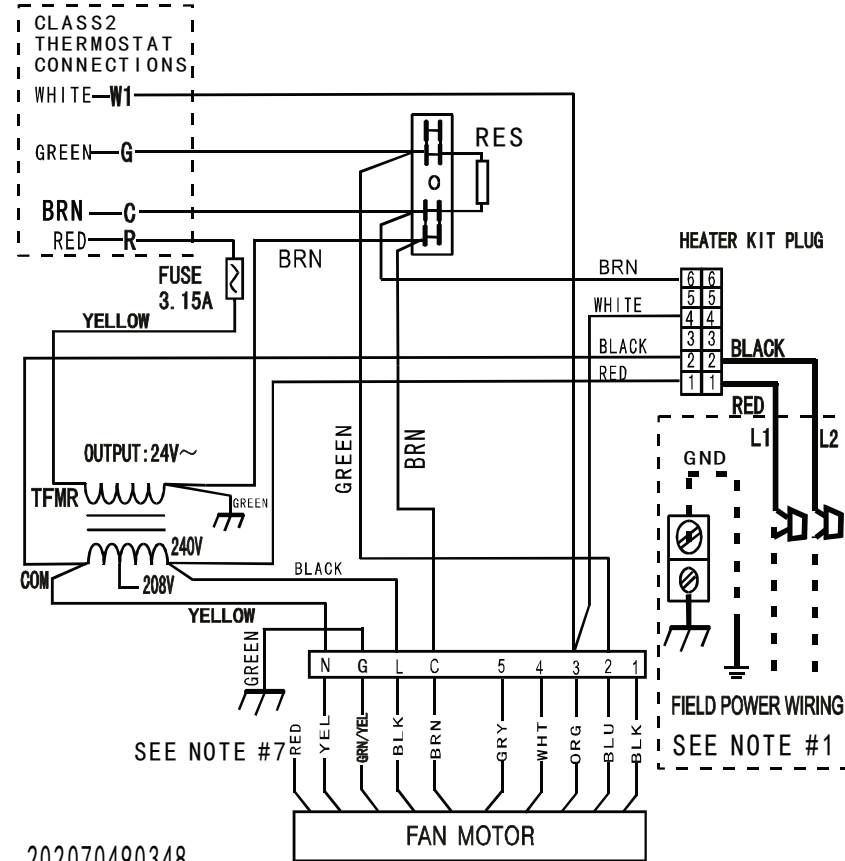
FIELD POWER WIRING

CAUTION:

NOT SUITABLE FOR USE ON SYSTEMS EXCEEDING 150V TO GROUND

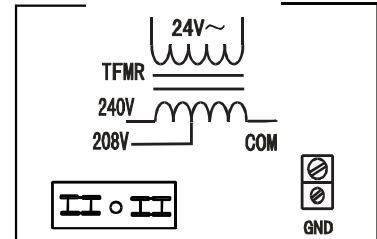
ATTENTION:

NE CONVIENT PAS AUX INSTALLATIONS DE PLUS DE 150V A LA TERRE



202070490348

COMPONENT ARRANGEMENT



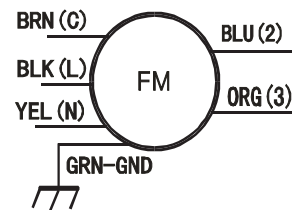
SPEED TAP SELECTION

- 1 LOW
- 2 MEDIUM LOW
- 3 MEDIUM
- 4 MEDIUM HIGH
- 5 HIGH

SEE NOTE #5, #6 & #8.

NOTES:

- 1: Use Copper Wire (75°C Min) Only Between Disconnect Switch And Unit .
- 2: To Be Wired In Accordance With NEC And Local Codes.
- 3: If Any Of The Original Wire ,As Supplied,Must Be Replaced.Use The Same Or Equivalent Type Wire.
- 4: Connect R To R,G To G,Etc.See Outdoor Instruction For Details.
- 5: To Change Speed Tap,Move Green Wire Desired Terminal.
- 6: See Airflow Tables For Tap Usage.
- 7:Factory Wires May Be Present,DO NOT USE.
- 8:Taps 2 & 4 Have a 90s Delay Off, Taps 1, 3 & 5 are 30s.



TFMR TRANSFORMER
FM FAN MOTOR
GND GROUND
RES RESISTOR
- - - FIELD POWER WIRING

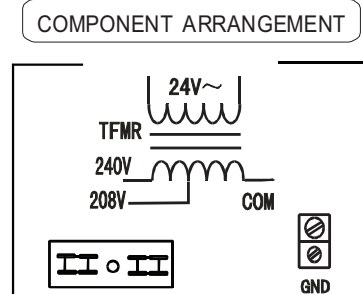
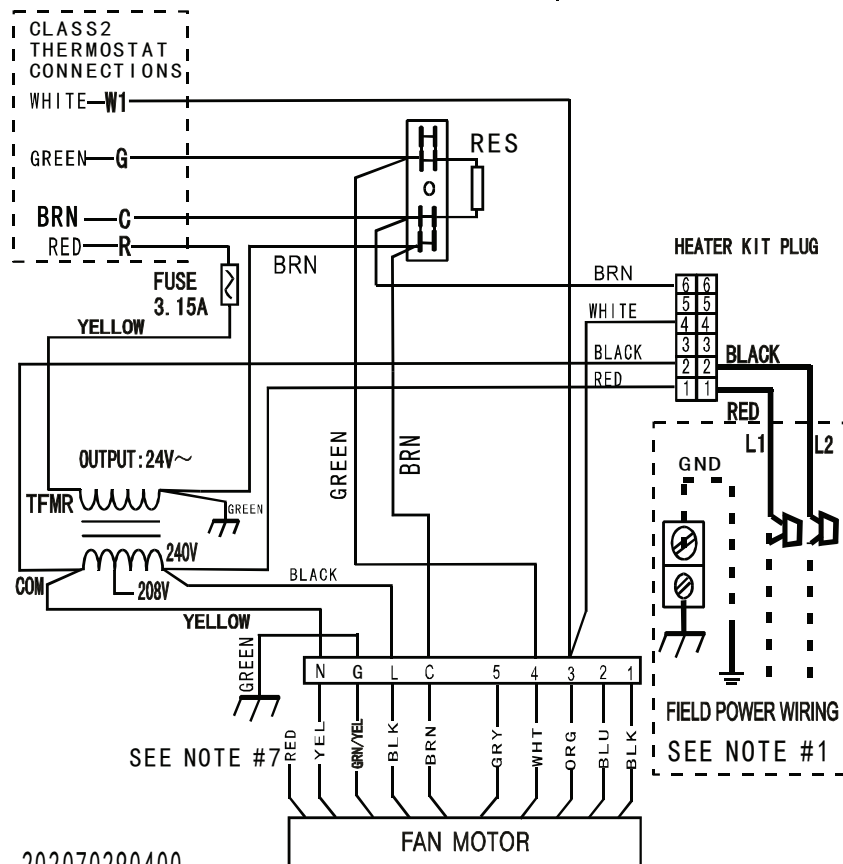
Fig. 25 - FFMANP(019, 031)

A14323

SEE RATING PLATE FOR VOLTS&HERTZ
FIELD POWER WIRING

CAUTION:
NOT SUITABLE FOR USE ON SYSTEMS EXCEEDING 150V TO GROUND

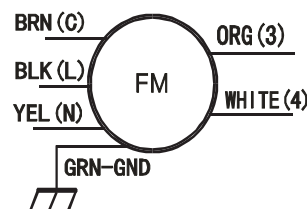
ATTENTION:
NE CONVIENT PAS AUX INSTALLATIONS DE PLUS DE 150V A LA TERRE



SPEED TAP SELECTION

- | | |
|---|-------------|
| 1 | LOW |
| 2 | MEDIUM LOW |
| 3 | MEDIUM |
| 4 | MEDIUM HIGH |
| 5 | HIGH |

SEE NOTE #5, #6 & #8.



NOTES:

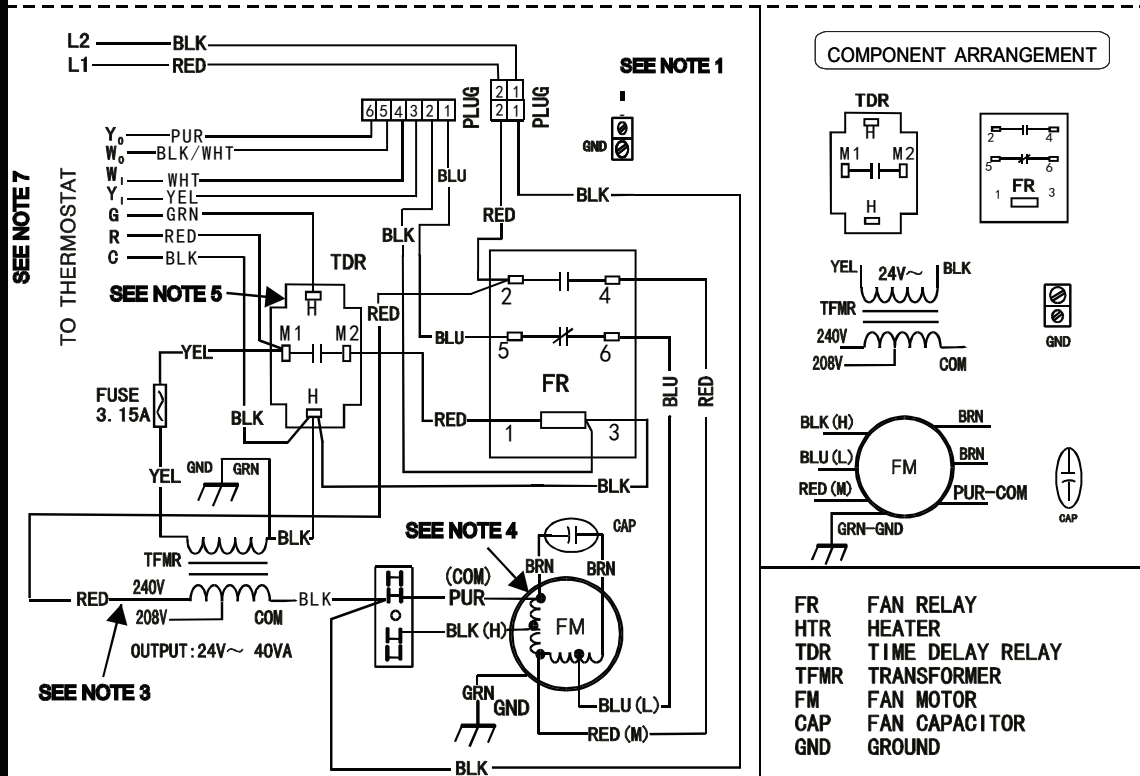
- 1: Use Copper Wire (75°C Min) Only Between Disconnect Switch And Unit .
- 2: To Be Wired In Accordance With NEC And Local Codes.
- 3: If Any Of The Original Wire ,As Supplied,Must Be Replaced.Use The Same Or Equivalent Type Wire.
- 4: Connect R To R,G To G,Etc.See Outdoor Instruction For Details.
- 5: To Change Speed Tap,Move Green Wire Desired Terminal.
- 6: See Airflow Tables For Tap Usage.
- 7:Factory Wires May Be Present,DO NOT USE.
- 8:Taps 2 & 4 Have a 90s Delay Off, Taps 1, 3 & 5 are 30s.

TFMR TRANSFORMER
FM FAN MOTOR
GND GROUND
RES RESISTOR
- - - FIELD POWER WIRING

Fig. 26 - FFMANP(025, 037)

SCHEMATIC DIAGRAM
SEE RATING PLATE FOR VOLTS&HERTZ
FIELD POWER WIRING

CAUTION:
 NOT SUITABLE FOR USE ON SYSTEMS EXCEEDING 150V TO GROUND
ATTENTION:
 NE CONVIENT PAS AUX INSTALLATIONS DE PLUS DE 150V ALA TERRE



NOTES:

- 1: Use copper wire(75°C min) only between disconnect switch and unit, To be wired in accordance with N.E.C. and local codes. Fan coils equipped with electric heater connect power supply to terminal block. Cooling controls wiring not used with electric heaters.
- 2: If any of the original wire as supplied must be replaced, use the same or equivalent type wire.
- 3: Remove the red lead from "240V" terminal and then connect the red lead to "208V" terminal on the transformer for 208 volts.
- 4: Factory default fan speed is Medium, FM red wire connected to FR #4; For HI speed connect FM black wire to FR #4; For LOW speed connect FM blue wire to FR #4, and FM red wire connected to FR #6. Always connect the unused FM wire to the dummy terminal block.
- 5: TDR has a 1-20s on delay when "G" is energized and a 50-70s off delay when "G" is de-energized.
- 6: Connect R to R, G to G, etc. See outdoor or indoor instructions for details.

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Fig. 27 - FPM(A,B)N(U,C)

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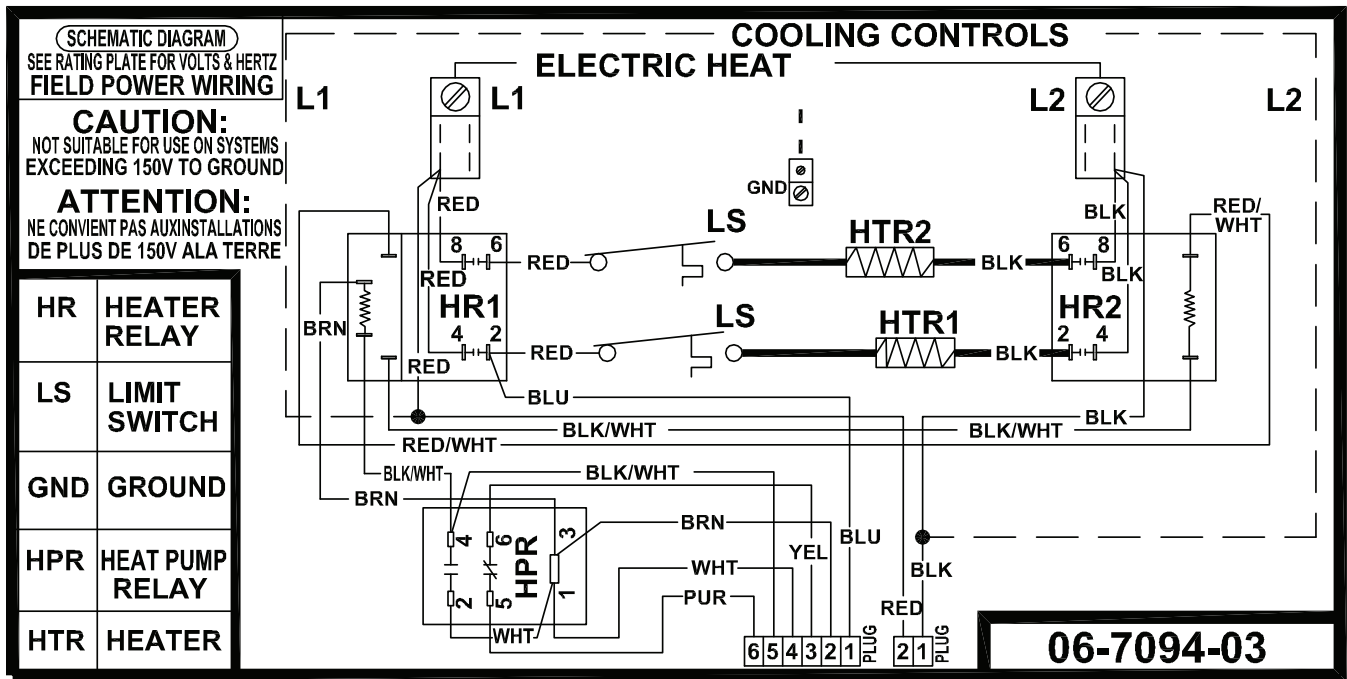


Fig. 28 - EHK3 With Heater Relays

A150101

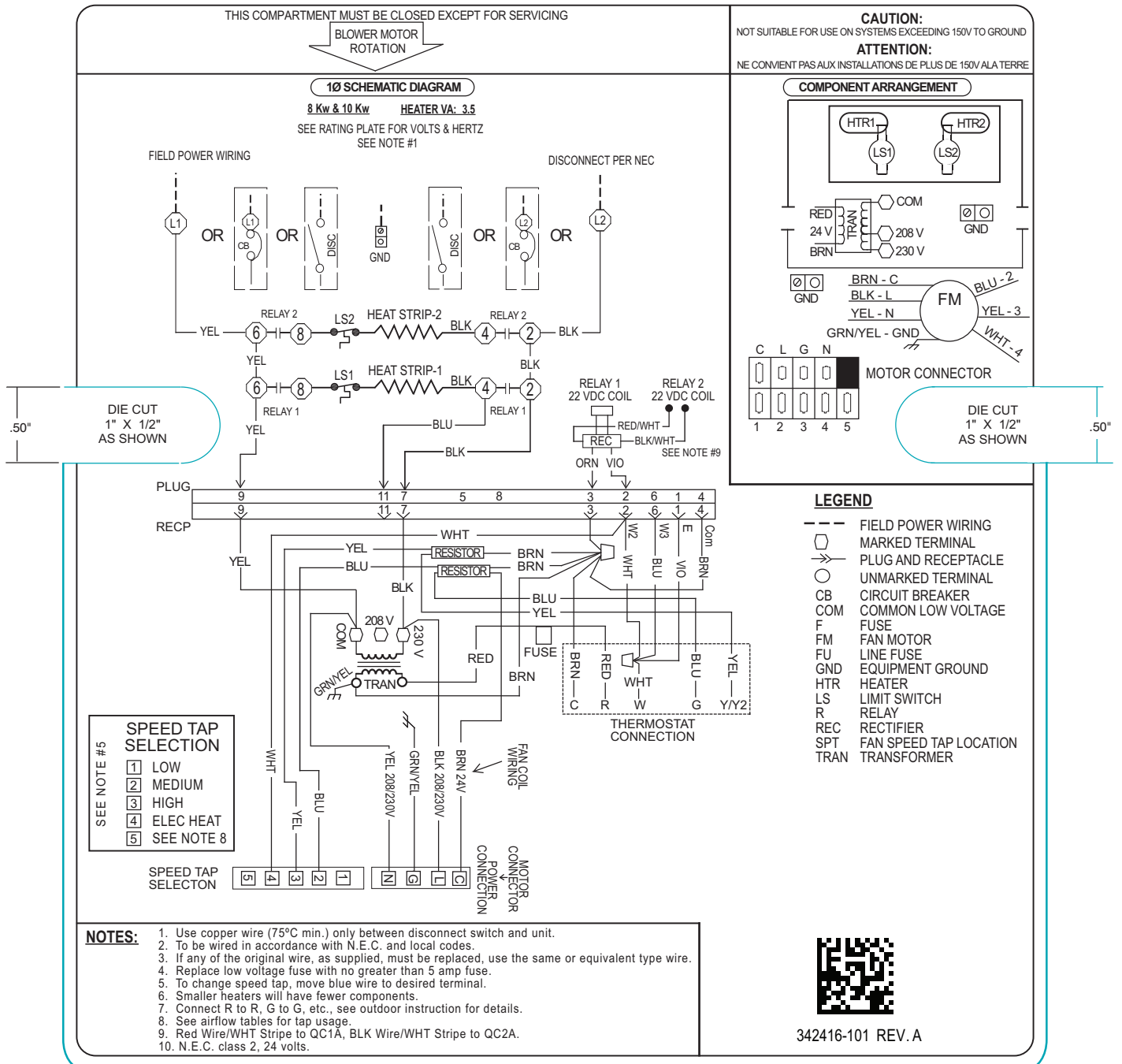


Fig. 29 - FZ4A

A160102

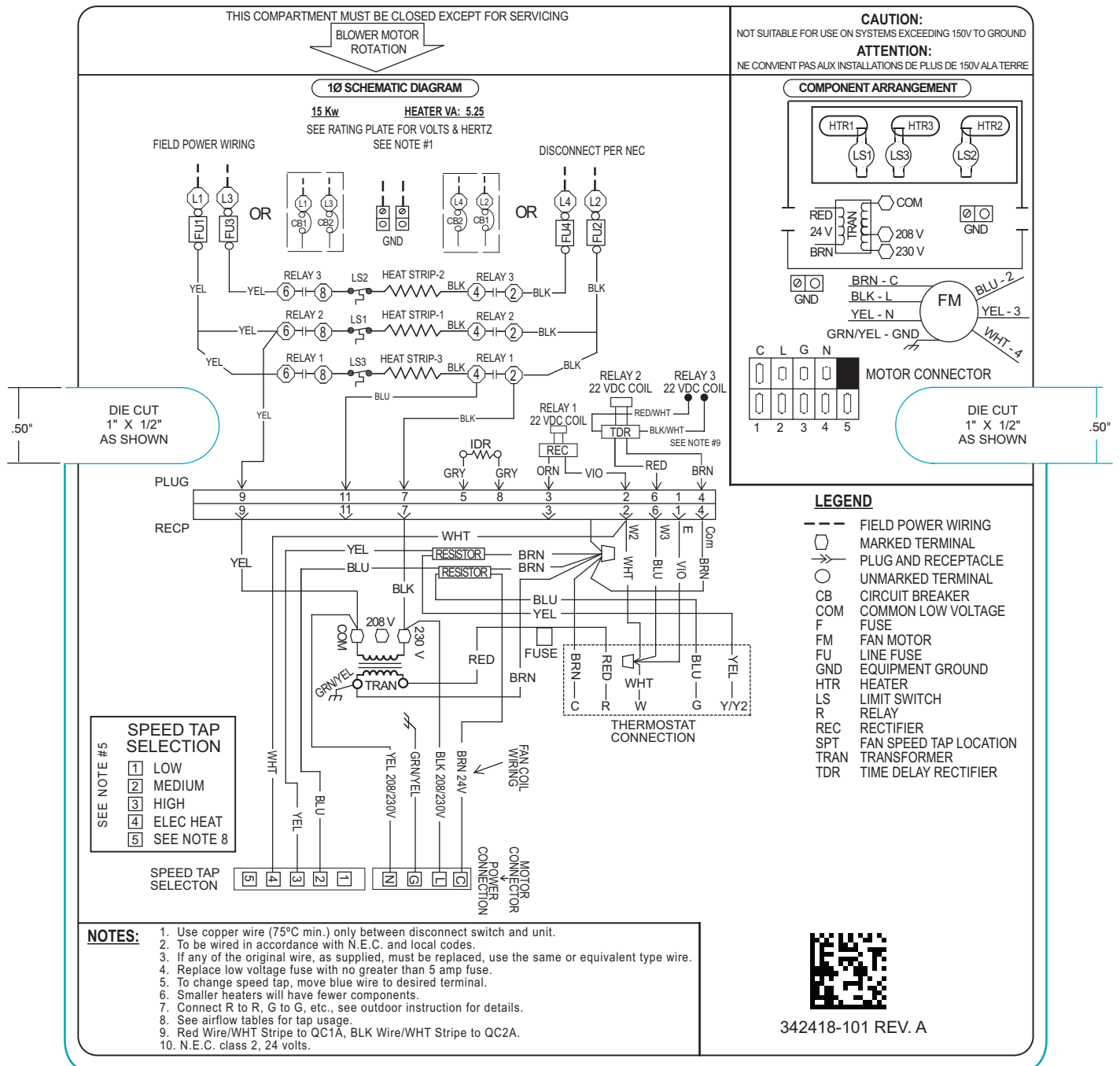


Fig. 30 - FZ4A

A160104

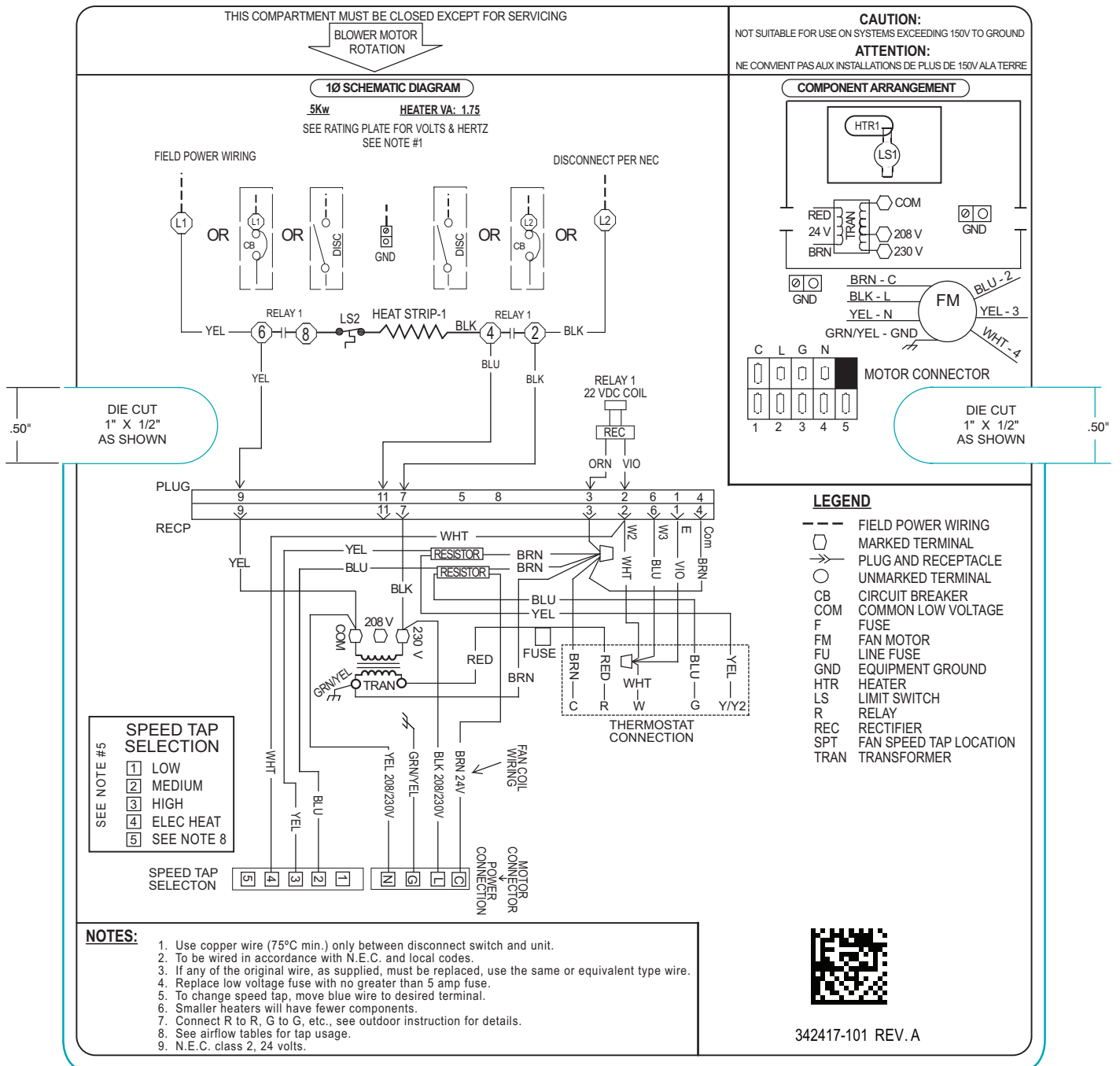


Fig. 31 - FZ4A

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